

Artefacts in single-cell mtDNA analyses misinform phylogenies

<https://doi.org/10.1038/s41586-026-10777-0>

Received: 24 August 2024

Accepted: 8 June 2026

Published online: 19 August 2026

 Check for updates

Caleb A. Lareau^{1✉}, Michael Spencer Chapman^{2,3}, Livius Penter⁴, Tal Nawy¹, Dana Pe'er^{1,5} & Leif S. Ludwig^{6,7,8✉}

ARISING FROM: C. Weng et al. *Nature* <https://doi.org/10.1038/s41586-024-07066-z> (2024).

The identification of somatic mutations through sequencing has enabled the reconstruction of clonal relationships in native human tissues. In a recent study, Weng et al.¹ introduced Regulatory Multiomics (transcriptomics and chromatin accessibility) with Deep Mitochondrial Mutation Profiling (ReDeeM), reporting unprecedented resolution of cellular phylogenies from single-cell mitochondrial DNA (mtDNA) variant data. However, we find that common sequencing artefacts substantially distort potential phylogenetic reconstructions, and removal of probable artificial variants yields markedly disparate phylogenetic inferences. Consequently, the study does not provide robust evidence that somatic mtDNA variation alone permits the confident inference of highly resolved cellular phylogenies.

Cellular barcoding to trace lineages and clonality has tremendous potential to resolve developmental processes, map dynamics of tissue regeneration and investigate mechanisms of disease onset. The sequencing-based identification of somatic mtDNA and nuclear DNA (nDNA) mutations has revealed clonal mosaicism in nearly all human tissues, such as in the haematopoietic system². Although mtDNA mutations have been utilized as effective markers of clonality, the unique properties of mitochondrial genetics have historically limited their utility for reconstructing more complex cellular phylogenies. Thus, it was with great interest that we read the report by Weng et al.¹, which claimed an unprecedented ability to connect individual cells with mtDNA mutations and resolve fine-grained phylogenies. The authors report an order-of-magnitude greater connectedness between cells via mtDNA mutations to resolve detailed phylogenetic trees—a capability that is not supported by contemporary results for clonal analysis or lineage tracing^{3–5} (Supplementary Discussion). Given the discrepancies between ReDeeM and related work, we revisited the methodology and results reported by Weng et al.¹.

ReDeeM introduces new experimental protocols and computational workflows for analysing mtDNA mutations from single-cell multiomics data. Experimentally, mtDNA is enriched from 10x Genomics single-cell multi-omic sequencing libraries and deeply sequenced, resulting in around five PCR copies (that is, individual sequencing reads) per mtDNA molecule. Together with bioinformatic mtDNA variant consensus calling, the technological advances include the depletion of sequencing errors, which can present a source of false-positive heteroplasmy in mtDNA analyses^{3–5}. However, the authors utilize new downstream workflows with insufficient orthogonal validation. ReDeeM requires only a single mtDNA molecule for a mutation to call a cell positive, despite sequencing these libraries beyond saturation. This is an unusual choice, as heteroplasmic allele frequency thresholds such as $\geq 10\%$

variant allele frequency (VAF) are customarily used to account for: (1) the multi-copy nature of the mitochondrial genome (150–600 copies per cell in blood); (2) the likelihood that low-VAF variants are lost during mitochondrial turnover; and (3) the asymmetric distribution of mitochondrial genomes during cell divisions⁶. Thus, a variant supported by a single mtDNA molecule may propagate in only one daughter cell and disappear from the other daughter cell lineage during cell division, limiting clonal or phylogenetic inference. Similarly, the unequal segregation of mitochondria during cell division means that variants present in two or three copies have a much higher chance of being lost to cellular lineage branches during division than those at high VAFs.

To determine the effect of low-support mtDNA variants, we reassessed the cell-by-variant count matrices of the four profiled donors. Despite saturated sequencing, 75.4–85.1% of mutation calls within individual cells were supported by only one sequenced mtDNA molecule—the minimal possible evidence, which would be considered insufficient in previous workflows (Extended Data Fig. 1a). Increasing the requirement to two molecules per cell causes a marked loss (87.8–99.9%) in connectedness in all four donors (Fig. 1a,b, Extended Data Fig. 1b and Supplementary Methods), indicating that the choice of binarizing at the unusual threshold of >0 drives the results obtained with ReDeeM. The authors require that at least two mtDNA variants be shared for connectedness between two cells. However, this choice is not justified over the common practice of establishing clonality based on as little as one high-confidence variant. Indeed, relaxing the requirement to a single shared mtDNA variant inexplicably increased the connectedness between cells reported by ReDeeM, to an average of 231 connections per cell (150–382 connections per cell per donor; Extended Data Fig. 1c). This corresponds to each cell sharing connections with 2.0–4.2% of sampled cells, which would reflect large clonal expansions that contradict reported haematopoietic architectures in young donors⁷. Thus, the parameters used by ReDeeM for establishing cell lineage (Fig. 1a and Extended Data Fig. 1b,c) deviate from best practices in the field, and minimal modifications result in markedly different quantitative conclusions.

To resolve the mtDNA mutations responsible for these observations, we summarized—per variant—the number of cells detected and the sequencing support, defined as the count of endogenous unique molecular identifiers (eUMIs)¹ per cell (Supplementary Methods). Analysis of these aggregations identified a set of mtDNA variants, mostly supported by one mtDNA molecule (mean per positive cell ≈ 1 ; threshold at 1.3 eUMIs), that were present in dozens to hundreds of cells across all donors. We term these low-mean and high-connectedness

¹Computational and Systems Biology Program, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ²Wellcome Sanger Institute, Hinxton, UK. ³Centre for Haemato-Oncology, Barts Cancer Institute, Queen Mary University of London, London, UK. ⁴Department of Hematology, Oncology, and Tumor Immunology, Campus Virchow Klinikum, Berlin, Charité–Universitätsmedizin Berlin, Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany. ⁵Howard Hughes Medical Institute, New York, NY, USA. ⁶Berlin Institute of Health at Charité–Universitätsmedizin Berlin, Berlin, Germany. ⁷Max-Delbrück-Center for Molecular Medicine in the Helmholtz Association (MDC), Institute for Medical Systems Biology (BIMSB), Berlin, Germany. ⁸Charité–Universitätsmedizin Berlin, Cluster of Excellence ImmunoPrecept, Berlin, Germany. ✉e-mail: lareauc@mskcc.org; leif.ludwig@bih-charite.de

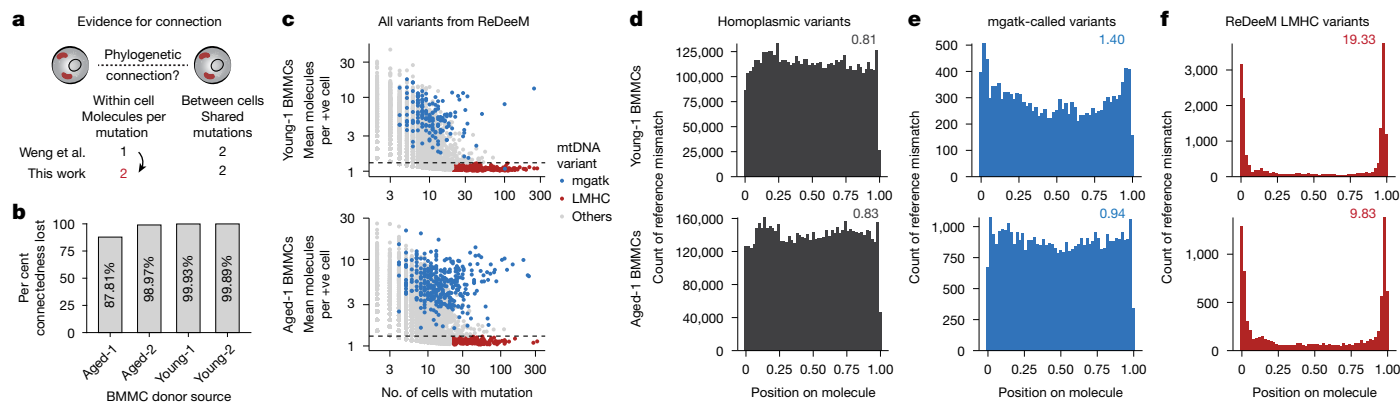


Fig. 1 | ReDeeM-based cell-cell connectivity is driven by low mean frequency but highly abundant variants enriched on the ends of molecules.

a, Schematic of key parameters to establish cell-cell connectedness. **b**, Connectedness lost for 4 donors after increasing the required mtDNA molecules from 1 to 2 for connectedness analysis. BMMC, bone marrow mononuclear cell. **c**, Annotation of variants from ReDeeM, noting LMHC

variants. **d–f**, Metapileup of mutation position on molecules for **(d)** homoplasmic, **(e)** mgatk-called and **(f)** ReDeeM-specific LMHC variants. For **d–f**, numbers above each plot represent the fold enrichment of variant mismatches at the molecule edges. Panel **a** adapted with permission from ref. 2, Elsevier.

(LMHC) variants (Fig. 1c, Extended Data Fig. 1d and Supplementary Methods). Although these variants comprise only 3.9–7.1% of all ReDeeM variants, their presence in many cells disproportionately affects cellular connectedness. Further, low-VAF mtDNA mutations are susceptible to loss during cell division or continuous mitochondrial turnover⁸, or to inadequate capture during library preparation, and are thus unreliable for phylogenetic reconstruction, as demonstrated by whole-genome sequencing reanalysis³ (Supplementary Discussion). Notably, established variant callers, including mgatk⁴, MQuad⁹ and GATK/Mutect2 (ref. 10) require stricter thresholds per individual mtDNA variant and do not identify LMHC variants (Fig. 1c, Extended Data Fig. 1d,e and Supplementary Methods).

As the reanalysis of ReDeeM data with established bioinformatics methods did not identify LMHC variants, we sought to define their origin. As previous work in next-generation sequencing data has identified artefacts at the edges of sequencing reads^{11,12}, including from duplex sequencing technologies with similar consensus error correction¹³, we examined the distribution of mismatched reference base calls underlying LMHC variants. As a control, we verified the approximately even distribution of maternally inherited homoplasmic mutations along sequenced molecules (Fig. 1d). This is consistent with near-random fragment generation by Tn5 tagmentation. Similarly, heteroplasmic somatic variants identified by mgatk⁴ and MQuad⁹ were distributed evenly (Fig. 1e and Extended Data Fig. 1d,e), even though their variant-calling heuristics do not explicitly consider mismatch position. In sharp contrast, the aggregated ReDeeM-derived LMHC variants were 10–20× enriched at the edges of mtDNA molecules in all 4 donors (Fig. 1f and Extended Data Fig. 1d). Further, 9 additional donors profiled via ReDeeM were also enriched 6.1–31.8-fold for reference mismatches at the edges of molecules (Extended Data Fig. 2a). Together, our analyses show that ReDeeM is unique in identifying LMHC variants that are supported by only one detected mtDNA molecule per cell, and these LMHC variants exhibit strong positional bias that resembles known sequencing artefacts^{11,12}.

Other sequencing technologies that use consensus calling explicitly filter edge artefacts or trim the edges of reads¹³, including for mtDNA sequencing¹⁴. Noting the pervasive accumulation of edge mismatches, we observed a similar edge-position bias in the original assay for transposase-accessible chromatin (ATAC) library as well as independent single-cell and bulk ATAC-seq libraries, from both mtDNA-derived and nuclear-derived accessible chromatin fragments (Extended Data Fig. 2b–d). These analyses indicate that spurious mismatches at the edges of sequenced DNA molecules are a technical feature of ATAC-seq

data (Supplementary Methods). To quantify this edge-position bias at the variant level, we developed a heuristic modelled after the gatk ReadPosRankSumTest¹⁵, which further reinforces the artificial nature of edge-enriched variants identified by ReDeeM (Supplementary Fig. 1; Supplementary Discussion).

To assess the downstream biological effect of low-support and probably artificial LMHC variants, we constructed dendrograms for the bone marrow mononuclear cell data from four donors using the weighted Jaccard distance and neighbour-joining algorithms proposed by Weng et al.¹ (Supplementary Methods). Trees reconstructed using the full ReDeeM variant set closely resembled the hierarchical structure in the original report¹ (Fig. 2a). We next assessed the degree of phylogenetic signal retained using variant-calling thresholds with greater biological confidence. By requiring two or more molecules of support per mtDNA variant, the revised trees exhibited a starkly dissimilar topology. This can be attributed to thousands of shared connections supported by variants only detected in a single molecule in the original trees (Fig. 2b). To quantify the degree by which the two trees share phylogenetic signal, we used a most recent common ancestor (MRCA) trio analysis² (Supplementary Methods) and found near-random relationships between the original tree and the tree requiring ≥ 2 eUMIs (33.8–39.2%; Fig. 2c, grey; random is 33.3%). Conversely, variants with exactly 1 eUMI retained a substantial fraction of MRCA trio concordance (64.7–68.8%; Fig. 2c, black), underscoring the divergent information content of LMHC variant calls compared with higher confidence mtDNA variants.

Following the original publication, the authors introduced ReDeeM filter-2, a new bioinformatics approach that explicitly trims the edges of assembled eUMIs. As no recommendations were made concerning when to use filter-1 or filter-2, we compared their performance and found that, despite removing only about 5% of (probably artefactual) mtDNA mutations, ReDeeM filter-2 reduces cell-cell connections by 53–99% compared with ReDeeM filter-1 (Extended Data Fig. 3). Consequently, cells in phylogenetic trees generated using ReDeeM filter-1 and filter-2 are positioned almost randomly relative to each other (Fig. 2d,e; 33.3% is random). This result reinforces a core principle in ReDeeM data: not all mutations contribute equally to connectivity and phylogenetic reconstruction. The vastly distinct phylogenies generated by the two ReDeeM methods further demonstrate the disproportionate effect of low-confidence mtDNA variants on phylogenetic inference.

The authors used a CRISPR-based lineage-tracing mouse model to validate phylogenetic inferences based on ReDeeM variants. However, our examination of these data revealed at least five undocumented experimental or computational differences from the human ReDeeM

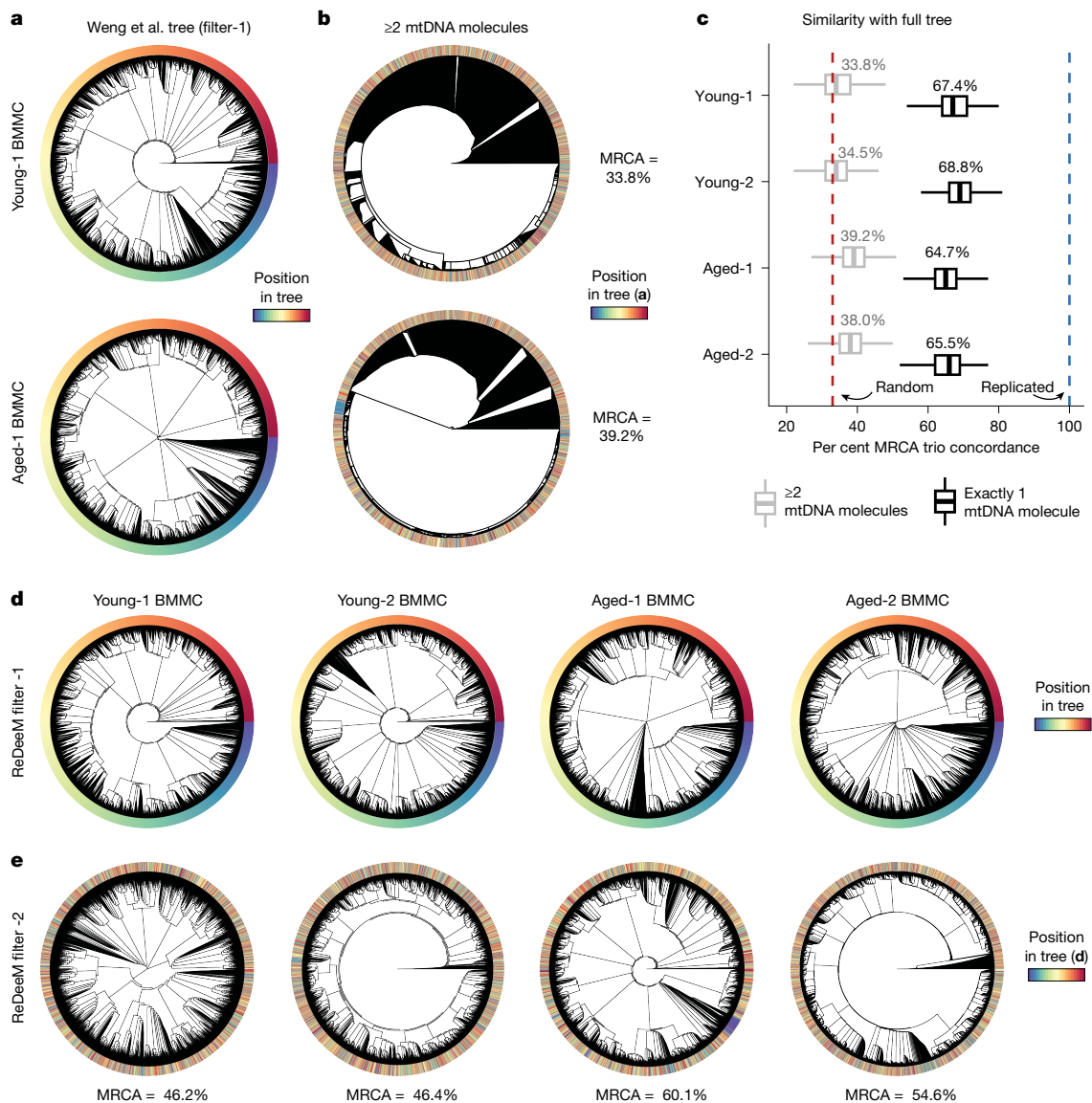


Fig. 2 | Subtle parameter choices yield highly divergent phylogenies. **a,b**, Phylogenetic trees of BMMCs were constructed using all variants from ReDeeM (**a**) or only variants that were supported by ≥ 2 mtDNA molecules (**b**). **c**, Quantification of tree similarity between **a** and **b** using trio analyses to

determine the MRCA. Box plots represent 100 bootstraps each of 1,000 trios using the tree in **a** as the 'truth' and the tree in **b** as the 'query'. **d,e**, Phylogenetic trees of four donors (columns) were constructed using ReDeeM filter-1 (**d**) and filter-2 that removes edge variants (**e**).

pipeline¹ (Supplementary Methods). Despite the fundamental differences in data composition and workflows, an accumulation of LMHC variants and edge bias of supporting molecules is also present in the mouse data (Supplementary Fig. 2). Other purported validations, such as the clonal k -nearest neighbour analyses, had independent logical inconsistencies that do not provide rigorous validation of the ReDeeM methodology (Supplementary Fig. 3 and Supplementary Discussion). Most importantly, we note that additional sources of confounding still limit the interpretation of ReDeeM data, particularly the high observed levels (estimated at over 11%) of ambient mtDNA contamination between cells during cell encapsulation, which would contribute to spurious phylogenetic inferences (Supplementary Fig. 4 and Supplementary Discussion). We further stress that although the phylogenetic trees between both ReDeeM filters are starkly discordant, neither phylogeny reported is likely to approximate the true cellular phylogeny. Specifically, in a synthetic donor mixing experiment, ReDeeM did not separate cells from two independent donors, which would otherwise be expected to show obviously distinct phylogenetic relationships if accurate (Supplementary Fig. 5).

Some innovations introduced by Weng et al.¹, including the targeted capture, deep sequencing and consensus calling of variants on mtDNA molecules, hold promise for establishing more sensitive single-cell mtDNA genotyping. However, our reanalysis reveals the existence of extensive technical artefacts upstream of library generation and sequence alignment, which cannot be corrected by PCR consensus calling and demand greater supporting evidence for confident mtDNA variant calls^{11,12}. Most mtDNA variants supported by exactly one molecule display a probably artificial edge-biased distribution, undermining their utility in downstream analyses, including phylogenetic reconstruction. These low-support artefacts, when called as variants, greatly exaggerate the cell–cell connectivity and strongly distort the inferred phylogeny, which is effectively random compared to variant-calling heuristics that require stricter thresholding (Fig. 2a–c). In response to these major technical concerns, the newly introduced ReDeeM filter-2 strategy mitigates the edge biases but produces a nearly random phylogeny relative to the original report¹ (Fig. 2d,e). The disparate connectivity metrics and resulting phylogenetic trees between the two filters reinforce that artefactual mtDNA variants were a substantial

Matters arising

driver of the purported phylogenetic signals in the original publication¹. Independent of these data, conceptually, low-VAF mtDNA variants are most susceptible to propagation in only one of the daughter cells or stochastic loss during mitochondrial turnover, thus generally undermining their biological utility for any type of phylogenetic or clonal inference. In sum, we strongly caution against the methods proposed by either ReDeeM workflow, particularly the reconstruction of human haematopoietic cell phylogenies (tree-building analyses) and connectedness inferences via mtDNA mutations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10777-0>.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data from this work were downloaded from GSE219015, which was previously published¹.

Code availability

Custom analysis code used to reproduce all inferences in this response is available at: <https://github.com/caleblareau/redeem-reanalysis-reproducibility>.

1. Weng, C. et al. Deciphering cell states and genealogies of human haematopoiesis. *Nature* **627**, 389–398 (2024).

2. Ludwig, L. S. et al. Lineage tracing in humans enabled by mitochondrial mutations and single-cell genomics. *Cell* **176**, 1325–1339.e22 (2019).
3. Chapman, M. et al. Mitochondrial mutation, drift and selection during human development and ageing. Preprint at *Research Square* <https://doi.org/10.21203/rs.3.rs-3083262/v1> (2023).
4. Lareau, C. A. et al. Massively parallel single-cell mitochondrial DNA genotyping and chromatin profiling. *Nat. Biotechnol.* **39**, 451–461 (2021).
5. Wang, X. et al. Clonal expansion dictates the efficacy of mitochondrial lineage tracing in single cells. *Genome Biol.* **26**, 70 (2025).
6. Roussou, R. et al. Real-time assessment of mitochondrial DNA heteroplasmy dynamics at the single-cell level. *EMBO J.* **43**, 5340–5359 (2024).
7. Mitchell, E. et al. Clonal dynamics of haematopoiesis across the human lifespan. *Nature* **606**, 343–350 (2022).
8. Chinnery, P. F. & Samuels, D. C. Relaxed replication of mtDNA: a model with implications for the expression of disease. *Am. J. Hum. Genet.* **64**, 1158–1165 (1999).
9. Kwok, A. W. C. et al. MQuad enables clonal substructure discovery using single cell mitochondrial variants. *Nat. Commun.* **13**, 1205 (2022).
10. Kim, M. et al. Single-cell mtDNA dynamics in tumors is driven by coregulation of nuclear and mitochondrial genomes. *Nat. Genet.* **56**, 889–899 (2024).
11. Pickrell, J. K., Gilad, Y. & Pritchard, J. K. Comment on ‘Widespread RNA and DNA sequence differences in the human transcriptome’. *Science* **335**, 1302 (2012).
12. Abascal, F. et al. Somatic mutation landscapes at single-molecule resolution. *Nature* **593**, 405–410 (2021).
13. Valentine, C. C. 3rd et al. Direct quantification of in vivo mutagenesis and carcinogenesis using duplex sequencing. *Proc. Natl Acad. Sci. USA* **117**, 33414–33425 (2020).
14. Serrano, I. M. et al. Mitochondrial haplotype and mito-nuclear matching drive somatic mutation and selection throughout ageing. *Nat. Ecol. Evol.* **8**, 1021–1034 (2024).
15. DePristo, M. A. et al. A framework for variation discovery and genotyping using next-generation DNA sequencing data. *Nat. Genet.* **43**, 491–498 (2011).

Author contributions C.A.L. led bioinformatics analyses with critical input and guidance from M.S.C., L.P., T.N., D.P. and L.S.L. All authors participated in the drafting and revising of the manuscript.

Competing interests C.A.L. and L.S.L. are named inventors on a patent related to mitochondrial lineage tracing (PCT/US2019/036583). D.P. is on the scientific advisory board of Insitro.

Additional information

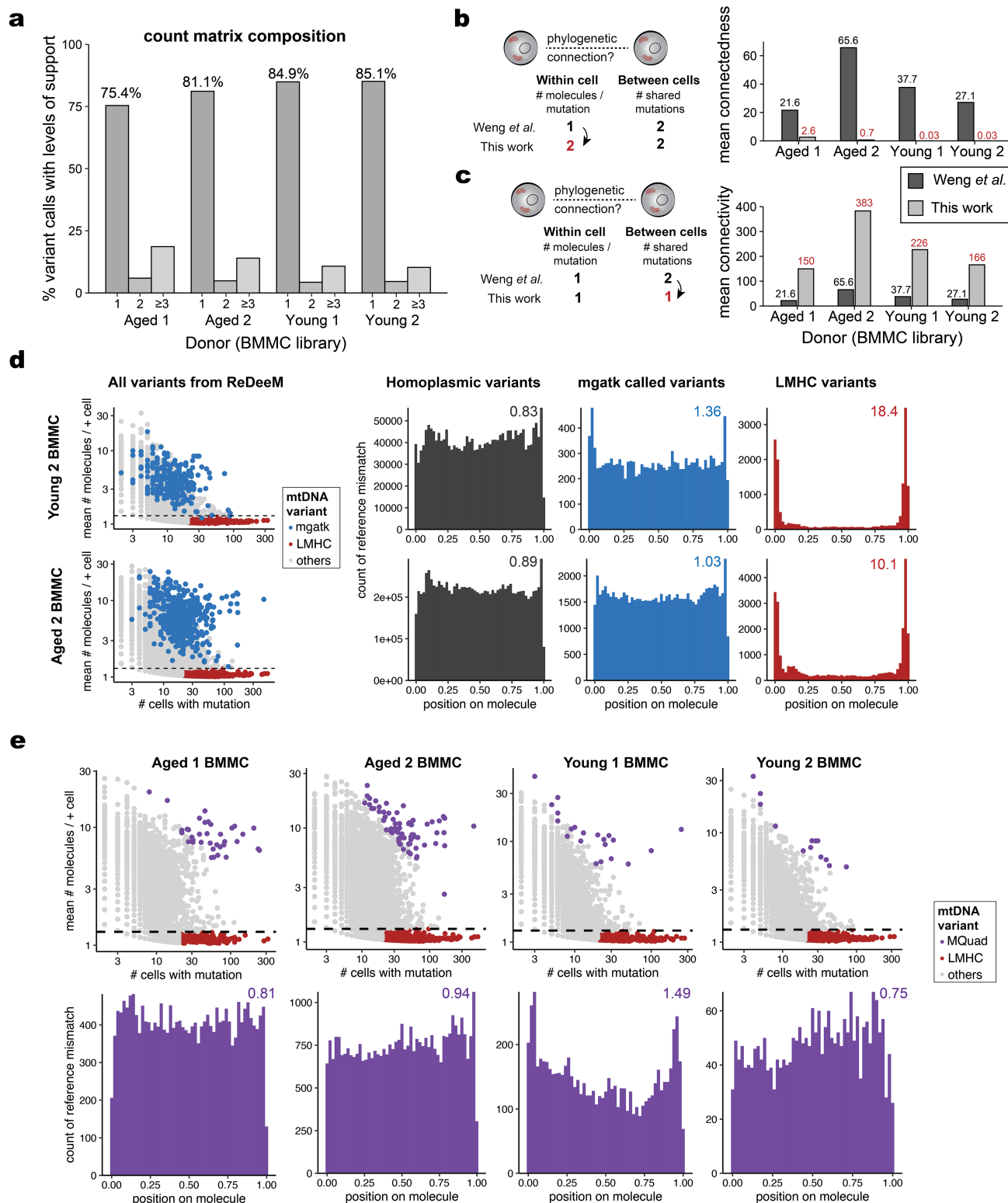
Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10777-0>.

Correspondence and requests for materials should be addressed to Caleb A. Lareau or Leif S. Ludwig.

Reprints and permissions information is available at <http://www.nature.com/reprints>.

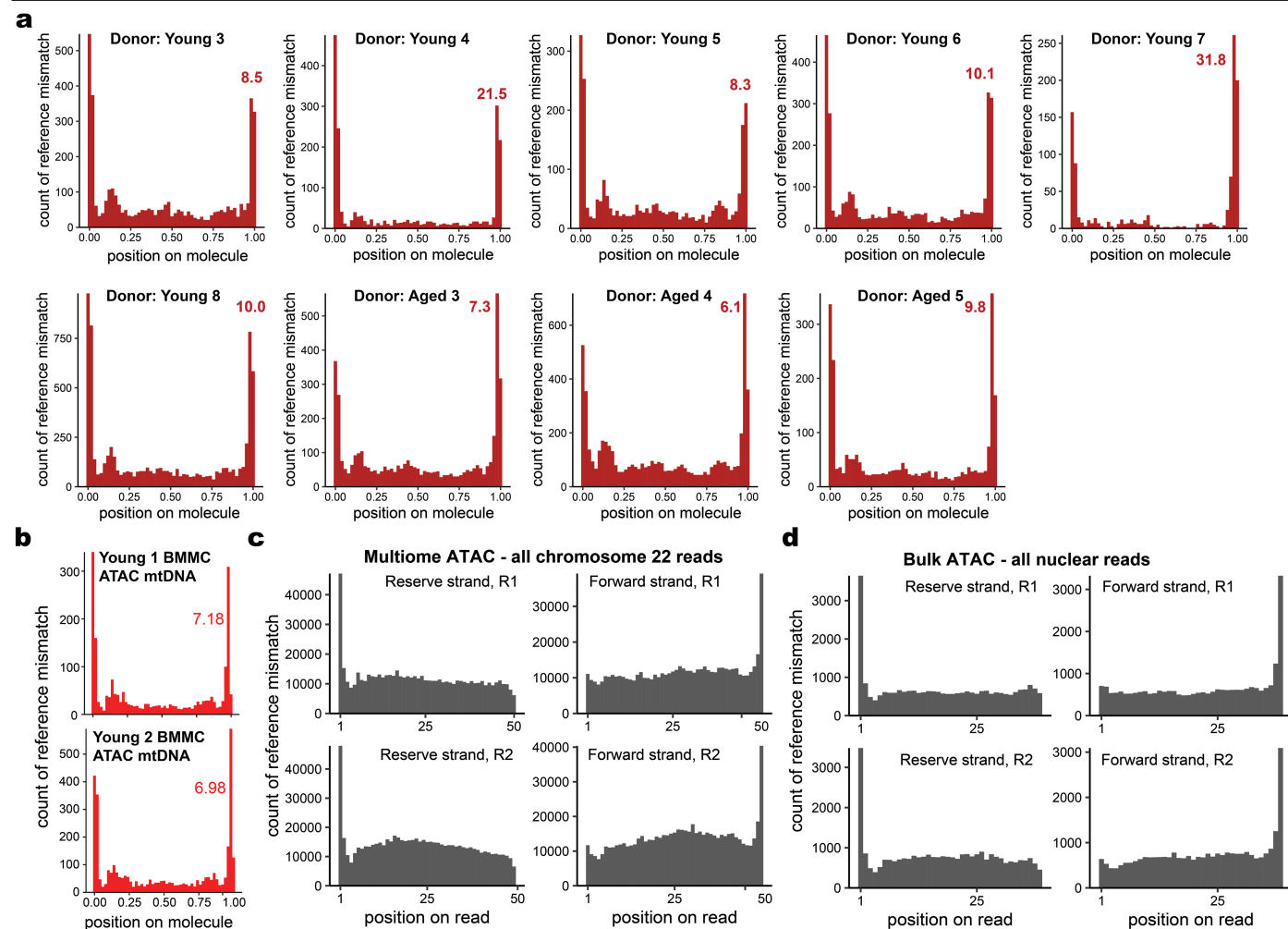
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature Limited 2026



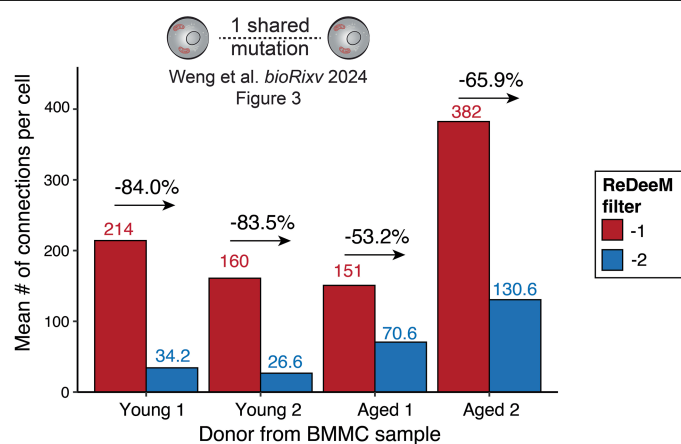
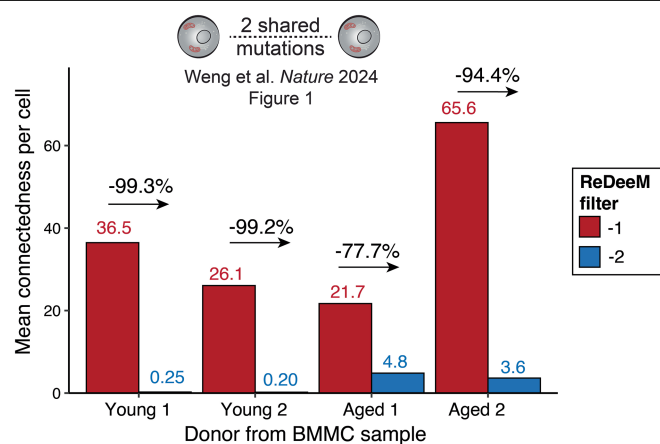
Extended Data Fig. 1 | The majority of ReDeeM-called variants have low support, and many represent artificial variant calls. (a) Summary of support per variant per cell, stratified by donor, represented as the percent of non-zero entries in the cell-by-variant ReDeeM matrix. The percent of variants with exactly 1, 2, or ≥ 3 molecules of support per cell appears above the corresponding bars. **(b,c)** Connectedness analyses, including key parameter adjustments (left) and the effects on mean connectivity (right). **(d)** The same

analyses as in Fig. 1, but for the two other donors profiled by Weng et al., showing similar positional bias in LMHC variants. **(e)** Annotation of all variants called using the ReDeeM workflow, including variants called by MQuad (purple) for four donors (top). Meta pileup of the position of the mismatches for variants called by MQuad (bottom). The values above each plot represent the fold enrichment of mismatches at the edges of molecules. Panels b,c adapted with permission from ref. 2, Elsevier.



Extended Data Fig. 2 | Mismatch position bias is present throughout ATAC-seq-derived molecules. (a) Pile-up of LMHC variants for 9 additional donors from the ReDeeM hashing experiments. Values in each panel represent the fold enrichment of mismatches at the edges of molecules relative to the center. (b) Pileup of LMHC variants from Fig. 1c and Extended Data Fig. 1d from the multiome ATAC-seq library. Values in each panel represent the fold enrichment of mismatches at the edges of molecules. (c) Pileup of all positions

containing a mismatch or reads mapping to the reverse strand or forward strand of chromosome 22, split by the sequencing read, from a publicly available single-nucleus peripheral blood mononuclear cell multiome experiment (Supplementary Methods). (d) Same as in (c) but for a bulk ATAC-seq experiment² summarizing all nuclear chromosomes. Note that no variant calling was performed for panels (c) and (d), such that pileups reflect all mismatches from the reference observed in the library.



Extended Data Fig. 3 | ReDeeM-based cell-cell connectivity is substantially reduced in both filter-1 and filter-2. Schematic of key parameters indicating the number of shared mutations required to establish connectivity between two cells, as previously applied by ReDeeM (top). The mean number of

connections per cell (y-axis) is shown with the relative differences between ReDeeM filters -1 and -2 being indicated in % across bone marrow mononuclear cells (BMBCs) from four profiled donors (bottom). Adapted with permission from ref. 2, Elsevier.